



Programming Interface to PostgreSQL

Object Explorer

Servers (1)

- PostgreSQL 18
 - Databases (1)
 - Login/Group Roles (21)
 - pg_checkpoint
 - pg_create_subscription
 - pg_database_owner
 - pg_execute_server_program
 - pg_maintain
 - pg_monitor
 - pg_read_all_data
 - pg_read_all_settings
 - pg_read_all_stats
 - pg_read_server_files
 - pg_signal_autovacuum_worker
 - pg_signal_backend
 - pg_stat_scan_tables
 - pg_use_reserved_connections
 - pg_write_all_data
 - pg_write_server_files
 - postgres
 - user1
 - user2
 - user3

postgres/postgres@PostgreSQL 18*

postgres/postgres@PostgreSQL 18

Group Role - Login/Group Roles

General Definition Privileges Membership Parameters Security SQL

Name: serviceAPI

Comments: serviceAPI

Information icon: ⓘ Question icon: ?

You are currently running version 9.11 of pgAdmin 4,

- pg_checkpoint
- pg_create_subscription
- pg_database_owner
- pg_execute_server_program
- pg_maintain
- pg_monitor
- pg_read_all_data
- pg_read_all_settings
- pg_read_all_stats
- pg_read_server_files
- pg_signal_autovacuum_worker
- pg_signal_backend
- pg_stat_scan_tables
- pg_use_reserved_connections
- pg_write_all_data
- pg_write_server_files
- postgres
- user1
- user2

Group Role - Login/Group Roles

General Definition Privileges Membership Parameters Security SQL

Password

.....

Account expires

YYYY-MM-DD hh:mm:ss



Please note that if you leave this field blank, then the password will never expire.

Connection limit

-1



You are currently running version 9.11 of pgAdmin 4,

Login Role - serviceapi



General Definition Privileges Membership Parameters Security SQL

Can login?



Superuser?



Create roles?



Create databases?



Inherit rights from the
parent roles?



Can initiate streaming
replication and
backups?



Bypass RLS?



Close

Reset

Save

General Definition Privileges Membership Parameters Security SQL

Member of



User/Role

WITH ADMIN

Members



User/Role

WITH ADMIN



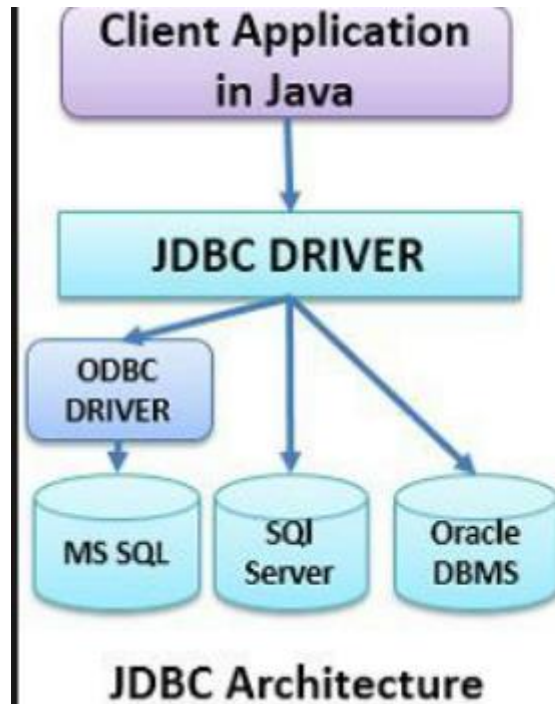
× Close

↺ Reset

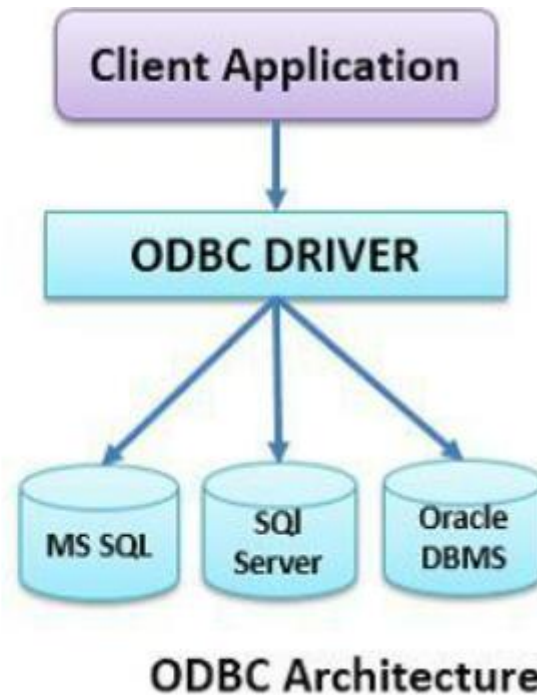
💾 Save

```
SELECT
  current_user,
  has_schema_privilege(
    current_user,
    'classicmodels',
    'USAGE'
  ) AS schema_usage,
  has_table_privilege(
    current_user,
    'classicmodels.accounts',
    'SELECT'
  ) AS can_select,
  has_table_privilege(
    current_user,
    'classicmodels.accounts',
    'UPDATE'
  ) AS can_update;
```

ODBC vs JDBC



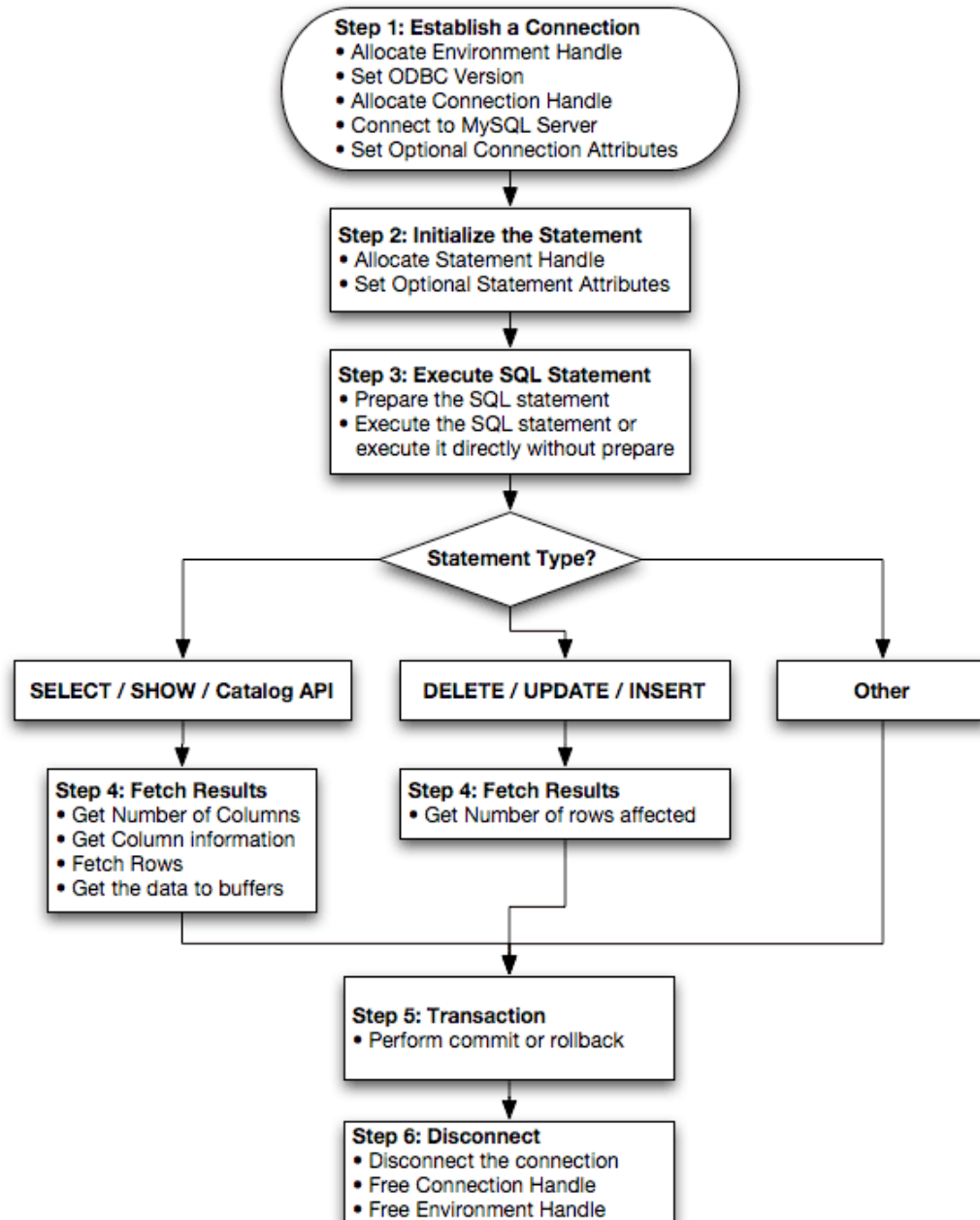
Vs



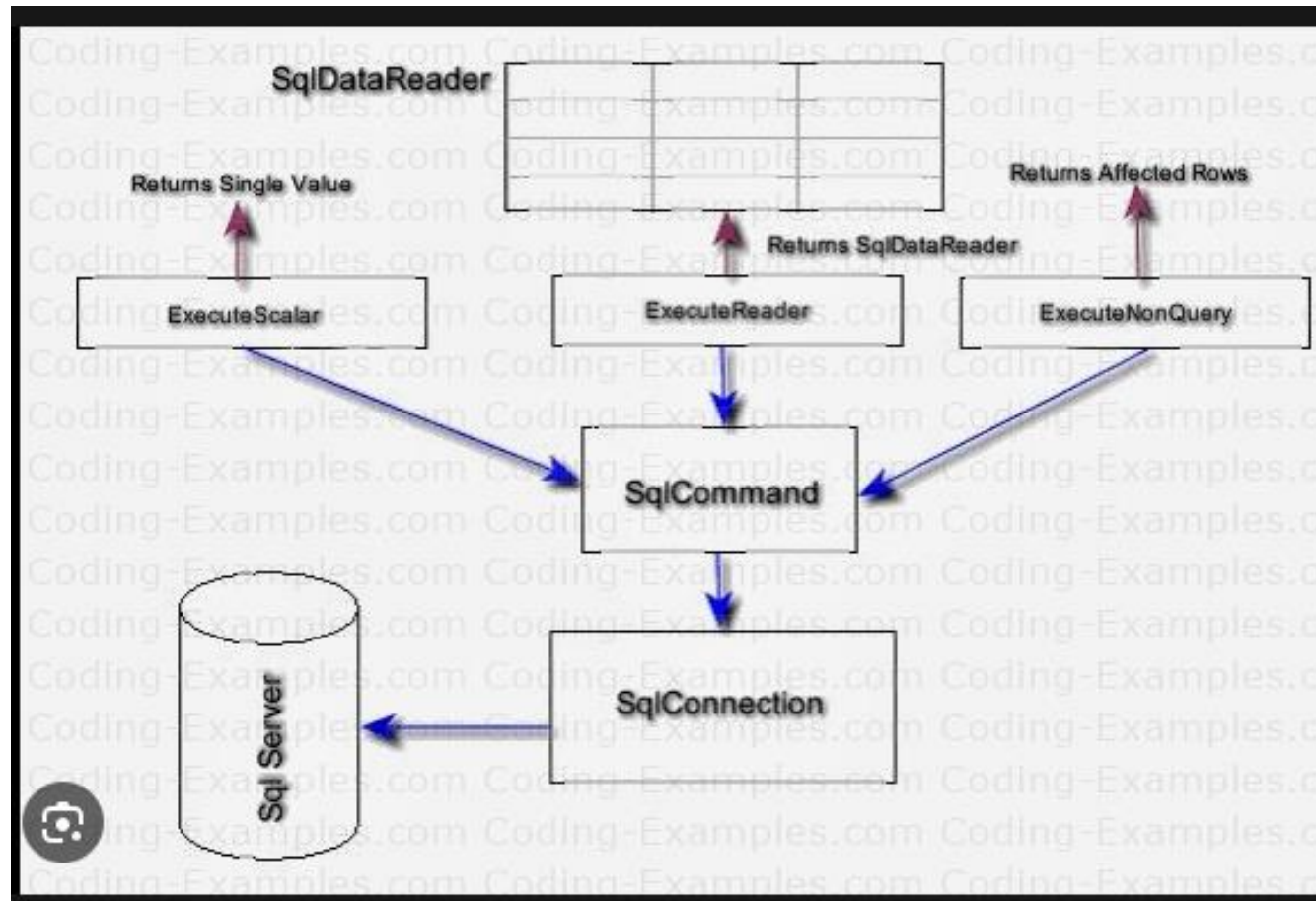
ODBC vs JDBC

	ODBC	JDBC
ชื่อเต็ม	Open Database Connectivity	Java Database Connectivity
ผู้กำหนดมาตรฐาน	Microsoft	Sun Microsystems (ปัจจุบัน Oracle)
เปิดตัว	1992	1997
ภาษาเจ้าของ	C / C++	Java
สถาปัตยกรรม	Native library (DLL/SO)	Java library (.jar)
เรียกจากภาษา	C, C++, C#, VB, Python, Delphi	Java (JVM) เท่านั้น
ไฟล์ไคลเอนต์	.dll (Windows), .so (Linux)	.jar
การติดตั้ง	ติดตั้งที่ OS + ตั้ง DSN	วาง .jar ใน classpath

Figure 6.1 Connector/ODBC Programming Flowchart

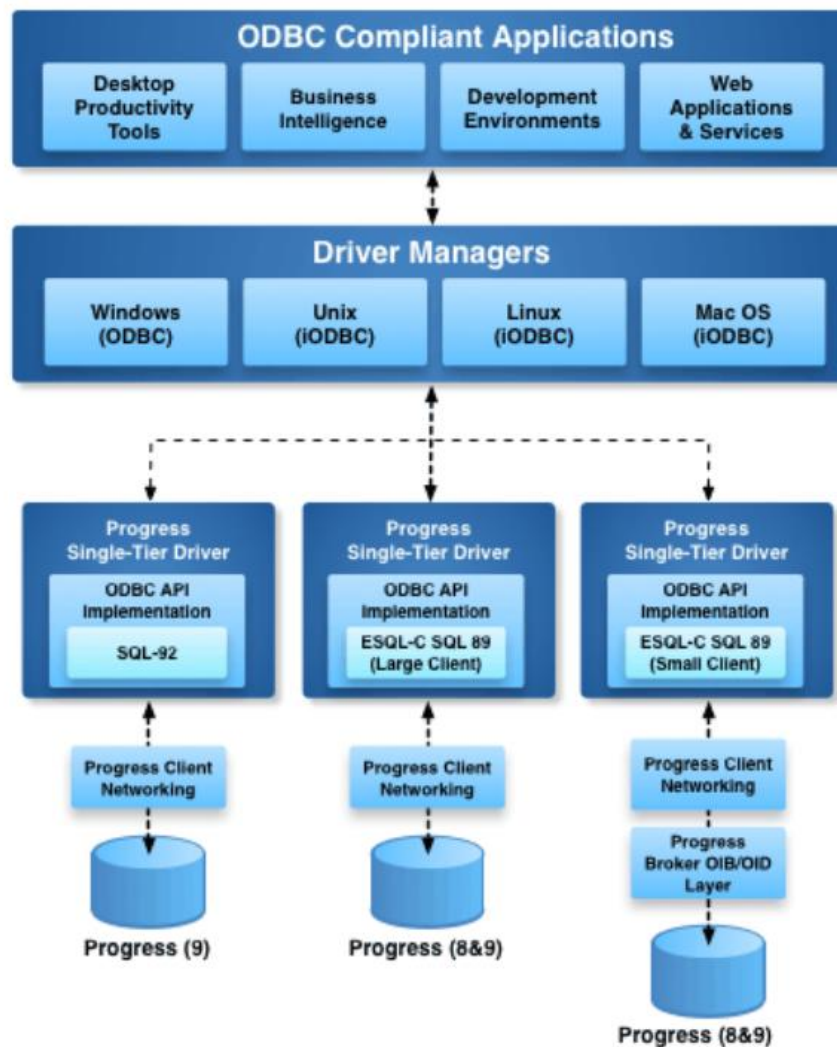


SQL execution command





.Net framework



https://www.postgresql.org/ftp/odbc/releases/REL-18_00_0004/

← → ↻ ⓘ postgresql.org/ftp/odbc/releases/REL-18_00_0004/

September 24, 2020: PostgreSQL 13 Beta 4 Released:

Quick Links

- Downloads
 - Packages
 - Source
- Software Catalogue
- File Browser

File Browser

Top → odbc → releases → REL-18_00_0004

Directories

- [Parent Directory]

Files

 psqlodbc-setup.exe	2026-09-14 10:16:24	8.2
 psqlodbc_arm64.msi	2026-09-14 10:16:25	4.1

Download history

🔍 Search download history

🏢 Your [browser is managed](#) by your organization

Today



psql

From



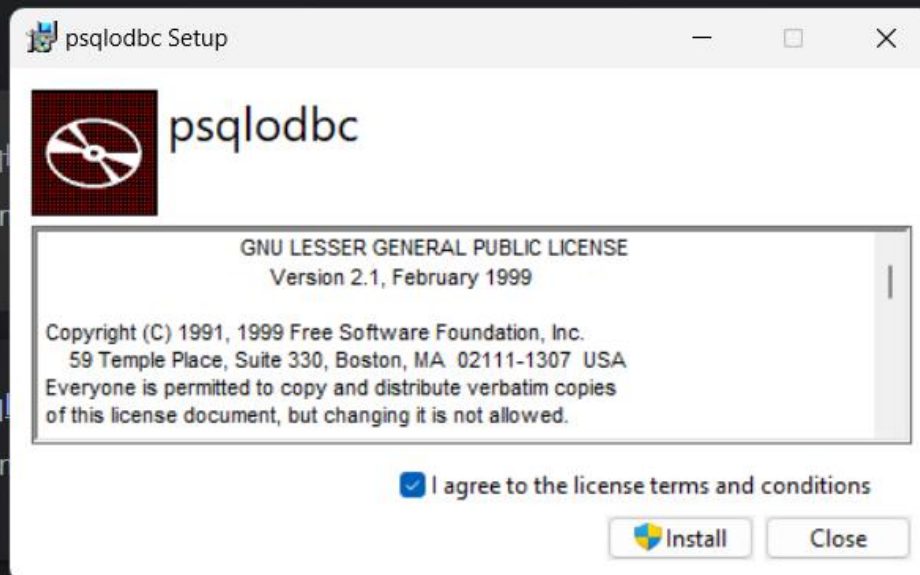
psql

From

September 22, 2026



ch16-2.pdf



Bind odbc drive กับ odbc manager

latest version of 17.14 to remain supported and receive the latest security and quality fixes or install Visual Studio 2026. [Update 17.14](#) [Install Visual Studio 2026](#)

App1

ConsoleApp1 Program

11 {
12 // Setup a connection string
13 // string szConnect
14 //
15 //
16 // Setup a connection string
17 string szConnect =
18 // "Driver={PostgreSQL Unicode};
19 "Driver={PostgreSQL Unicode};
20 "Server=localhost;" +
21 "Port=5432;" +
22 "SSLmode=disable;" +
23 "Database=postgres;" +
24 "Uid=serviceapi;" +
25 "Pwd=serviceapi;" ;
26
27 // Attempt to open a connection
28 OdbcConnection cnDB = new OdbcConnection(szConnect);
29
30 // The following code demonstrates how to catch & report an ODBC exception.
31 // To keep things simple, this is the only exception handling in this example.
32 // Note: The ODBC data provider requests ODBC3 from the driver. At the time of
33 // writing, the psqlODBC driver only supports ODBC2.5 - this will cause
34 // an additional error, but will *not* throw an exception.
35 try

ODBC Data Source Administrator (64-bit)

User DSN System DSN File DSN Drivers Tracing Connection Pooling About

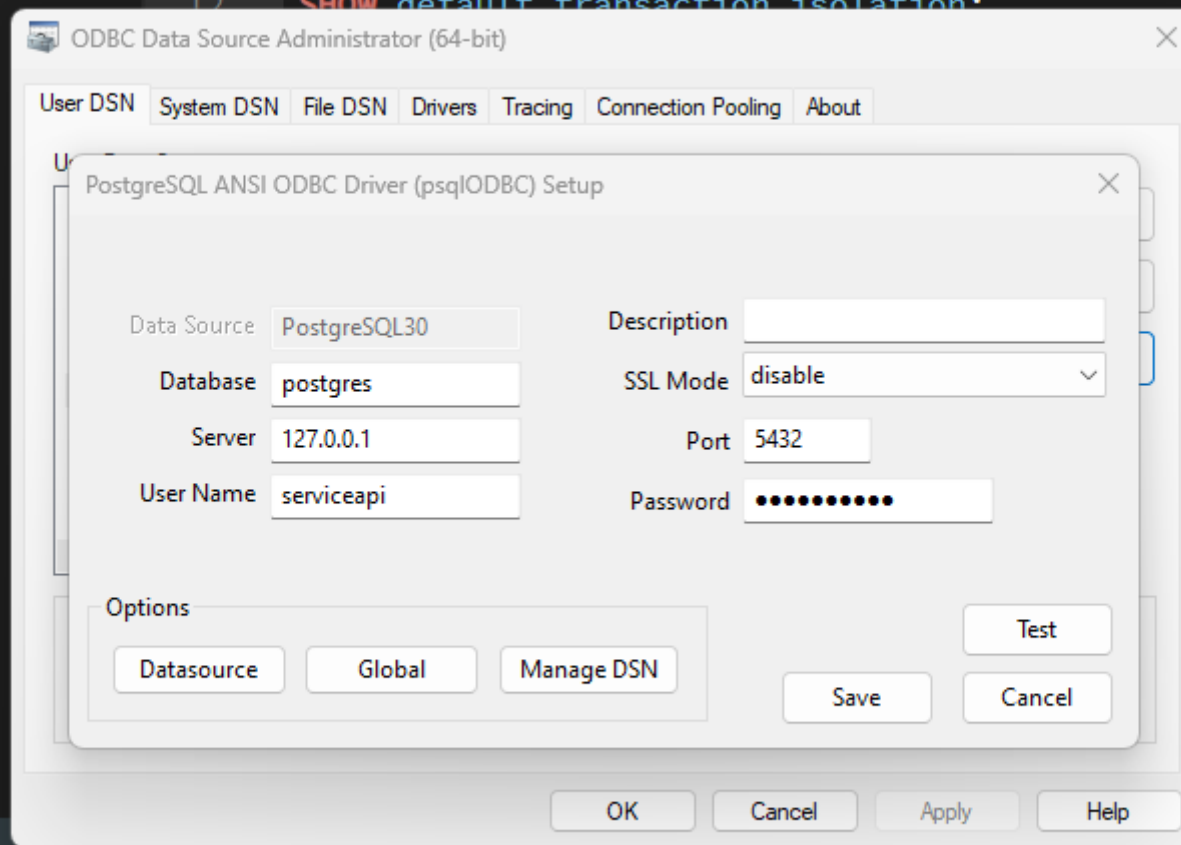
User Data Sources:

Name	Platform	Driver
dBASE Files	64-bit	Microsoft Access dBASE Driver (*.dbf, *.ndx, *.xls, *.xlsx, *.xlsm, *.xlt, *.xltm, *.xltm, *.xltm)
Excel Files	64-bit	Microsoft Excel Driver (*.xls, *.xlsx, *.xlsm, *.xlt, *.xltm, *.xltm, *.xltm)
MS Access Database	64-bit	Microsoft Access Driver (*.mdb, *.accdb)
MySql	32/64-bit	MySQL ODBC 8.0 Unicode Driver
PostgreSQL30	64-bit	PostgreSQL ANSI(x64)
PostgreSQL35W	32/64-bit	PostgreSQL Unicode
Visio Database Samples	64-bit	Microsoft Access Driver (*.mdb, *.accdb)



An ODBC User data source stores information about how to connect to the indicated data provider. A User data source is only visible to you and can only be used on this computer.

OK Cancel Apply Help



https://www

ChatGPT Classic

Insert Draw Design

Layout

Reset

New Slide

Section

Slides

Code editor with C# code snippets and a sidebar with a list of code files.

Create a new project

Recent project templates

Console App (.NET Framework) C#

Console App C++

ASP.NET Web Application (.NET Framework) C#

ASP.NET Core Web App (Model-View-Controller) C#

Dynamic-Link Library (DLL) C++

Shared Items Project C++

Android Bindings Library (Xamarin) C#

Android App (Xamarin) C#

Android Wear App (Xamarin) C#

ASP.NET Core Web App (Razor Pages) C#

Search for templates (Alt+S)

Clear all

C# Windows Web

APIs published as native AOT.

C# Linux macOS Windows API Cloud Service

Web Web API

MSTest Playwright Test Project New

A project that contains Playwright tests using MSTest test framework, that can run on .NET on Windows, Linux and MacOS.

C# Linux macOS Windows Desktop MSTest

Playwright Test Web

JUnit Playwright Test Project New

A project that contains Playwright tests using NUnit test framework, that can run on .NET on Windows, Linux and MacOS.

C# Linux macOS Windows Desktop NUnit Playwright

Test Web

ASP.NET Core Web App (Razor Pages)

A project template for creating an ASP.NET Core application with example ASP.NET Core Razor Pages content

C# Linux macOS Windows Cloud Service Web

Back Next

Configure your new project

Console App (.NET Framework)

C#

Windows

Console

Project name

ConsoleApp1

Location

D:\mydoc\COS3103\ppt.1.69\dotnet.code\

...

Solution name ⓘ

ConsoleApp1

☐

Place solution and project in the same directory

Framework

.NET Framework 4.7.2

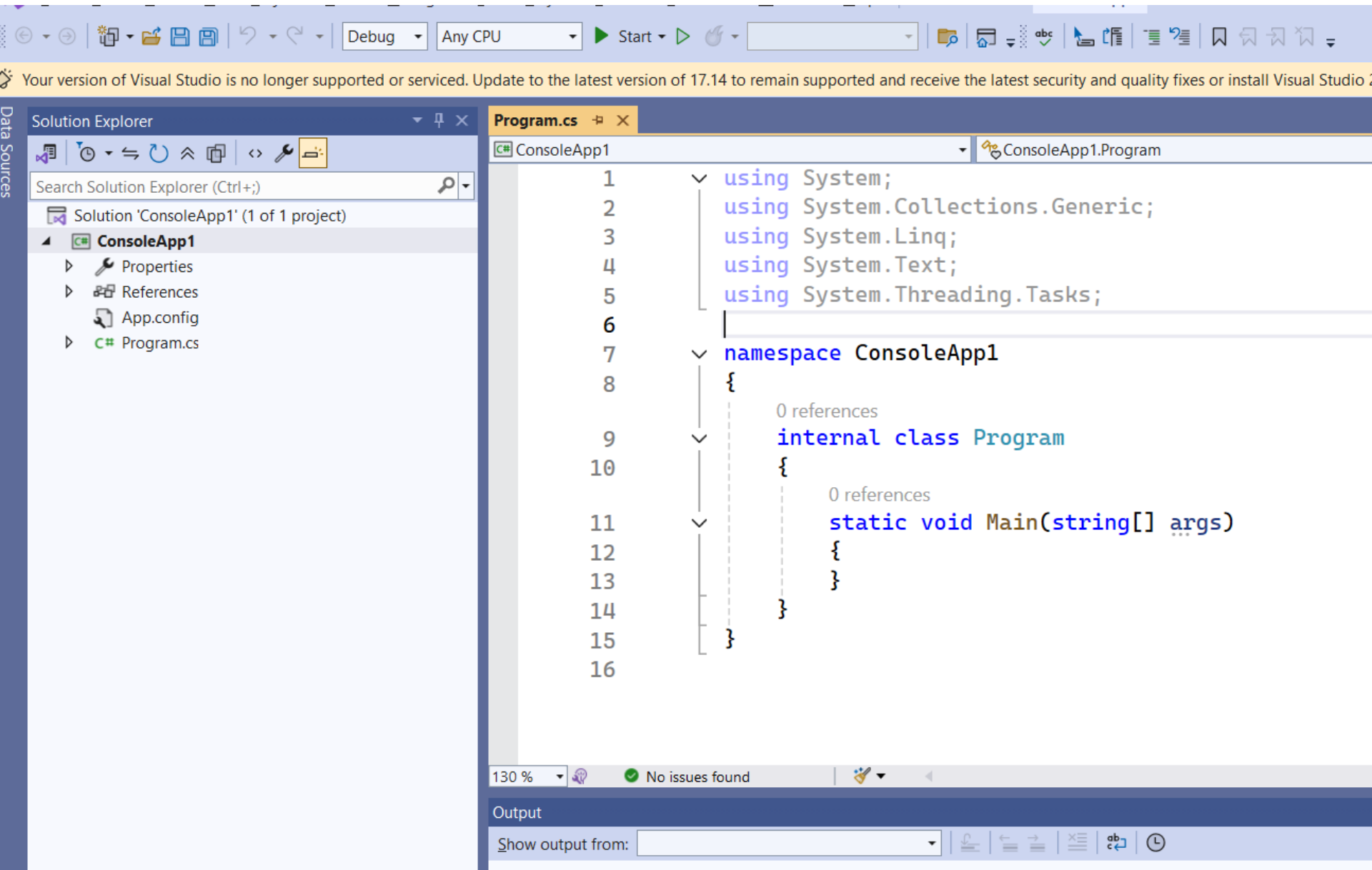
Project will be created in "D:\mydoc\COS3103\ppt.1.69\dotnet.code\ConsoleApp1\ConsoleApp1\"

Back

Create

<#>

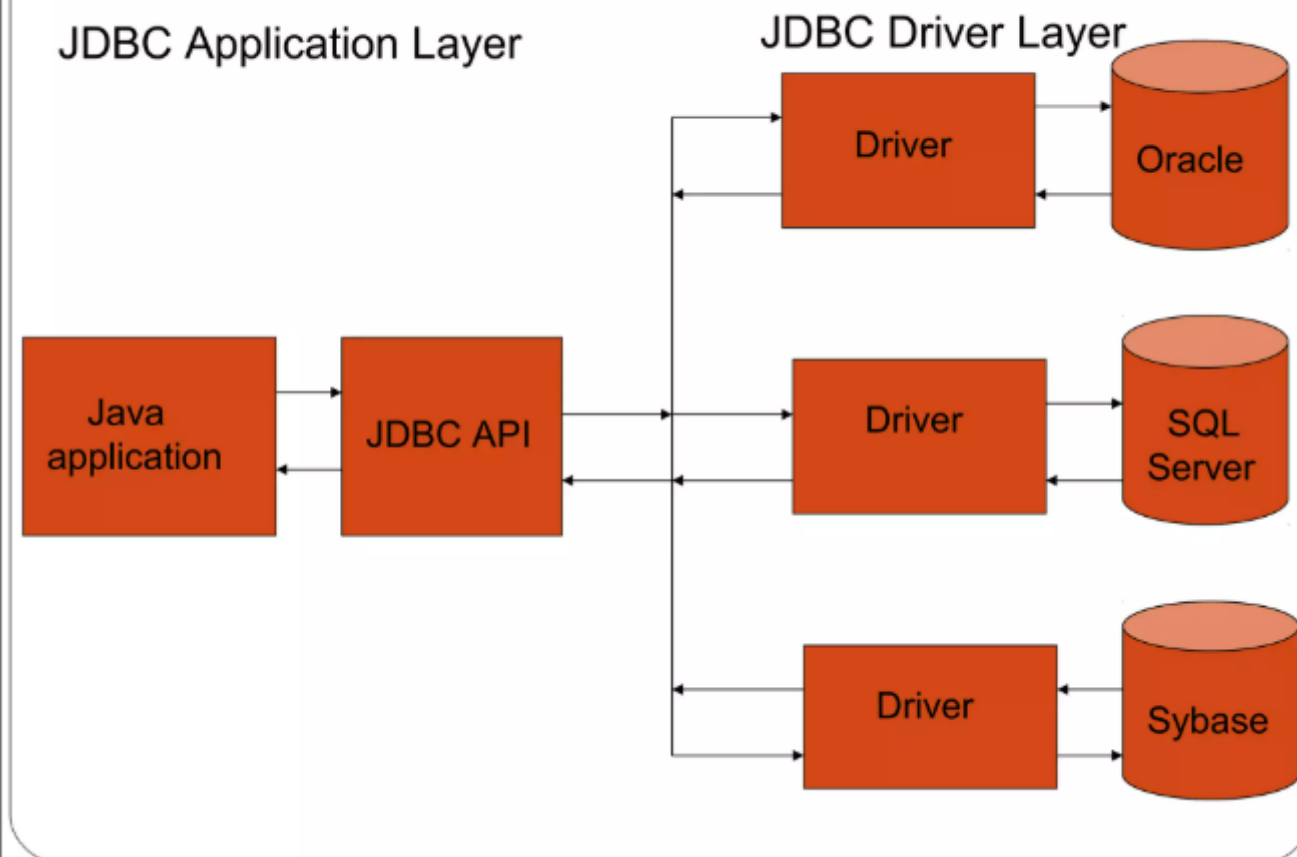
https://github.com/rapirakgmail/COS3103_169/tree/main/cshare_conn_db



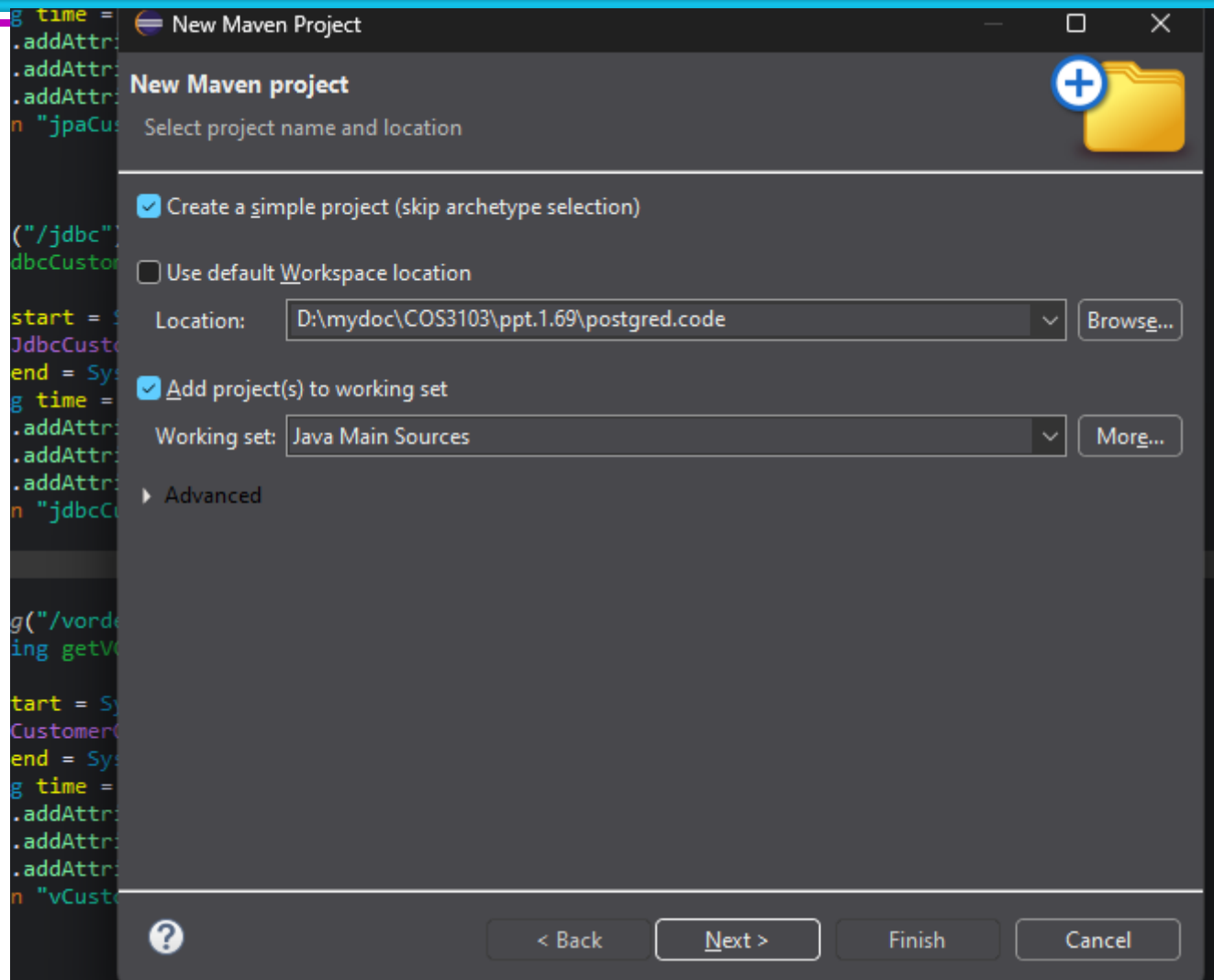


JAVA JDBC

The following figure shows the JDBC architecture:



Create maven project



mydemo [boot] [devtools]

```
47 {
48     long start = System.nanoTime();
49     List<JpaCustomerOrderCountDTO> l= mCustomerService.getJpaCustomerOrder();
50     long end = System.nanoTime();
51     String time =
52     model.addAttribute
53     model.addAttribute
54     model.addAttribute
55     return "jpaCus
56
57 }
58
59 @GetMapping("/jdbc")
60 String getjdbcCusto
61
62     long start =
63     List<JdbcCusto
64     long end = Sys
65     String time =
66     model.addAttribute
67     model.addAttribute
68     model.addAttribute
69     return "jdbcC
70
71 }
72
73 @GetMapping("/vord
74 public String getV
75 {
76     long start = Sy
77     List<Customer
78     long end = Sys
79     String time =
80     model.addAttribute
81     model.addAttribute
82     model.addAttribute
83     return "vCusto
84
85 }
86
87
88
89 }
```

New Maven Project

New Maven project

Configure project

Artifact

Group Id: app

Artifact Id: postgres-app

Version: 0.0.1-SNAPSHOT

Packaging: jar

Name:

Description:

Parent Project

Group Id:

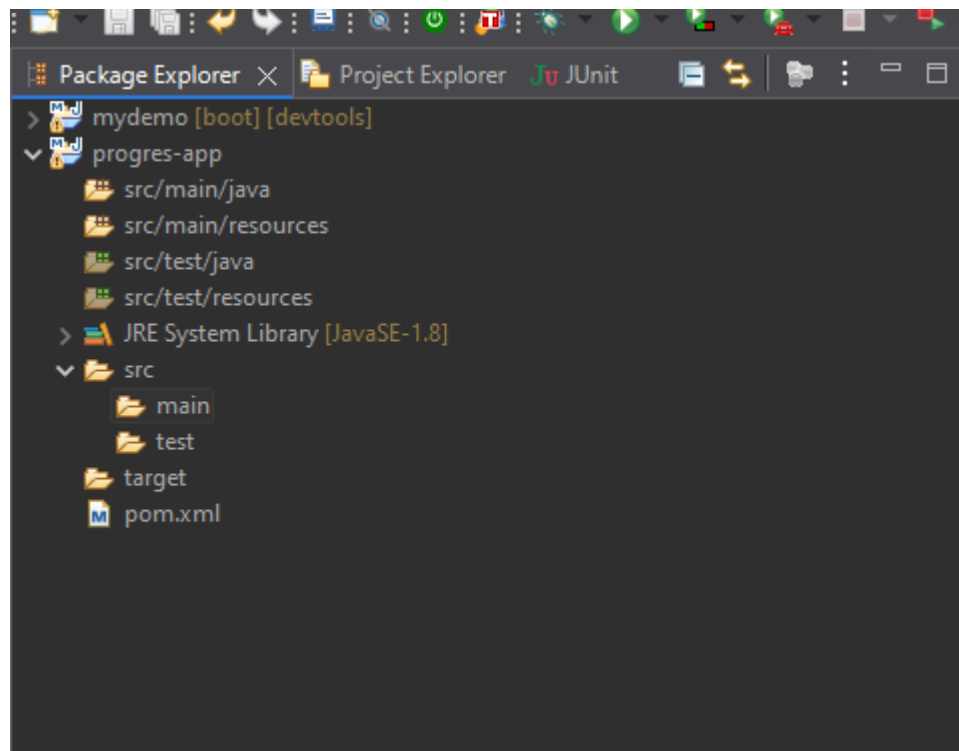
Artifact Id:

Version:

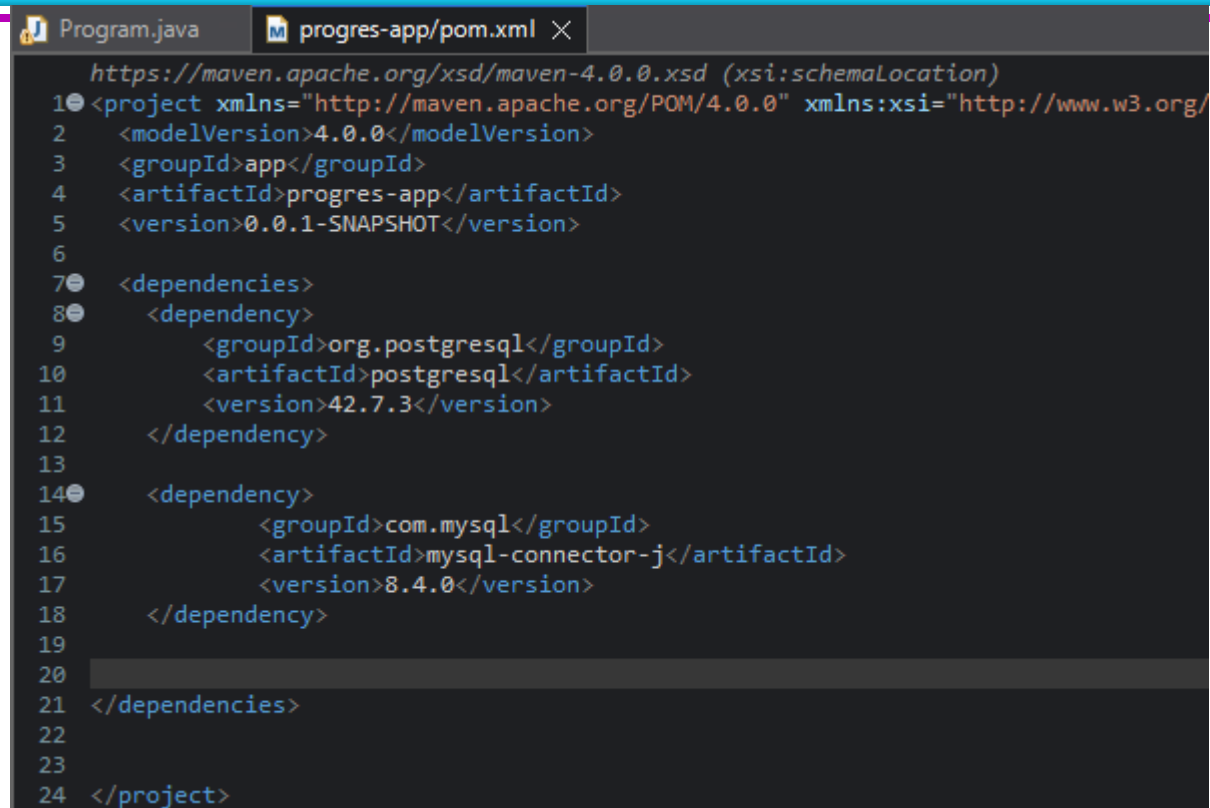
Browse... Clear

Advanced

< Back Next > Finish Cancel



Edit pom file



```
Program.java  progres-app/pom.xml X
https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/
2   <modelVersion>4.0.0</modelVersion>
3   <groupId>app</groupId>
4   <artifactId>progres-app</artifactId>
5   <version>0.0.1-SNAPSHOT</version>
6
7   <dependencies>
8     <dependency>
9       <groupId>org.postgresql</groupId>
10      <artifactId>postgresql</artifactId>
11      <version>42.7.3</version>
12    </dependency>
13
14    <dependency>
15      <groupId>com.mysql</groupId>
16      <artifactId>mysql-connector-j</artifactId>
17      <version>8.4.0</version>
18    </dependency>
19
20
21 </dependencies>
22
23
24 </project>
```

Java Class

Create a new Java class.

Source folder:

Package:

☐ Enclosing type:

Name:

Modifiers: ☐ public ☒ package ☐ private ☐ protected
☐ abstract ☐ final ☐ static

Superclass:

Interfaces:

Which method stubs would you like to create?

☒ public static void main(String[] args)
☐ Constructors from superclass
☒ Inherited abstract methods

Do you want to add comments? (Configure templates and default value [here](#))

☐ Generate comments

```
Program.java
1 package app;
2
3 import java.sql.*;
4 import java.util.ArrayList;
5 import java.util.List;
6
7 public class Program {
8
9     // =====
10    // DTO
11    // =====
12    public static class SaleDetailDTO {
13        public int employeeNumber;
14        public String lastName;
15        public String firstName;
16        public int customerNumber;
17        public String customerName;
18        public String salesRepEmployeeNumber;
19    }
20
21
22    private static Connection getConnection() throws SQLException {
23        Connection conn = DriverManager.getConnection("jdbc:postgresql://localhost:5432/postgres?sslmode=disable"
24            "serviceapi",//USER,
25            "serviceapi"//PASS
26        );
27        return conn;
28    }
29
30    private static Connection getConnectionWithSchema() throws SQLException {
31        Connection conn = DriverManager.getConnection("jdbc:postgresql://localhost:5432/postgres?sslmode=disable"
32            "serviceapi",//USER,
33            "serviceapi"//PASS
34        );
35        try (Statement st = conn.createStatement()) {
36            st.execute("SET search_path TO classicmodels");
37        }
38        return conn;
39    }
40 }
```

JPA vs JDBC

